

Supplementary Information: Reviving, reproducing, and revisiting Axelrod's second tournament

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Overview. In this paper, we focus on reproducing Axelrod's second tournament [1]. The **Supplementary Information** provides additional information and results that complement the main text. Namely, it is organized as follows.

- **Section S1** lists all 63 original submissions, including each strategy's author and the name of its Fortran function in the `Axelrod_fortran` library.
- **Section S2** reports how the original strategies rank across multiple tournaments considered in the paper, including our faithful reproduction, noisy variants, the Stewart and Plotkin tournament, and `Axelrod-Python` tournaments.
- **Section S3** provides additional detail on the reproduction, including the corrected Fortran source code of *Champion* (`k61r`) and a reproduction of Axelrod's linear model.
- **Section S4** provides the complete roster and ranks for the Stewart and Plotkin tournament [2], which now includes all original strategies.
- **Section S5** points to the full list of strategies and ranks for the comprehensive `Axelrod-Python` tournament (including the original submissions), available online.
- **Section S6** presents the expanded `Axelrod-Python` tournament under 1% and 5% noise.
- **Section S7** shows the original Fortran source code of `k42r` and summarizes its behavior.
- **Section S8** lists all 270 original submissions and `Axelrod-Python` library strategies, including their rank in each of the new tournaments.
- **Section S9** includes plots of the interaction distributions, in terms of cooperation rates and payoffs, for both the original tournament and the expanded `Axelrod-Python` tournament.
- **Section S10** presents the results of the tournament under probabilistic match endings, which more closely reflect the intended stopping rule of Axelrod's second tournament.

S1 List of strategies from Axelrod's second tournament

Here we provide a complete list of the 63 strategies that participated in Axelrod's second tournament. The list includes the originally reported rank of each strategy, its author, and the name of its Fortran function implementation in the `Axelrod_fortran` library.

- | | |
|--|---|
| 1. k31r - Original rank: 23. Authored by Gail Grisell | 22. k52r - Original rank: 48. Authored by David A Smith |
| 2. k32r - Original rank: 10. Authored by Charles Kluepfel | 23. k53r - Original rank: 45. Authored by Henry Nussbacher |
| 3. k33r - Original rank: 59. Authored by Harold Rabbie | 24. k54r - Original rank: 58. Authored by William H Robertson |
| 4. k34r - Original rank: 52. Authored by James W Friedman | 25. k55r - Original rank: 42. Authored by Steve Newman |
| 5. k35r - Original rank: 11. Authored by Abraham Getzler | 26. k56r - Original rank: 38. Authored by Stanley F Quayle |
| 6. k36r - Original rank: 63. Authored by Roger Hotz | 27. k57r - Original rank: 31. Authored by Rudy Nydegger |
| 7. k37r - Original rank: 37. Authored by George Lefevre | 28. k58r - Original rank: 21. Authored by Glen Rowsam |
| 8. k38r - Original rank: 34. Authored by Nelson Weiderman | 29. k59r - Original rank: 40. Authored by Leslie Downing |
| 9. k39r - Original rank: 25. Authored by Tom Almy | 30. k60r - Original rank: 6. Authored by Jim Graaskamp and Ken Katzen |
| 10. k40r - Original rank: 35. Authored by Robert Adams | 31. k61r - Original rank: 2. Authored by Danny C Champion |
| 11. k41r - Original rank: 7. Authored by Herb Weiner | 32. k62r - Original rank: 51. Authored by Howard R Hollander |
| 12. k42r - Original rank: 3. Authored by Otto Borufsen | 33. k63r - Original rank: 57. Authored by George Duisman |
| 13. k43r - Original rank: 39. Authored by R D Anderson | 34. k64r - Original rank: 17. Authored by Brian Yamachi |
| 14. k44r - Original rank: 5. Authored by William Adams | 35. k65r - Original rank: 47. Authored by Mark F Batell |
| 15. k45r - Original rank: 50. Authored by Michael F McGurrin | 36. k66r - Original rank: 20. Authored by Ray Mikkelsen |
| 16. k46r - Original rank: 14. Authored by Graham J Eatherley | 37. k67r - Original rank: 27. Authored by Craig Feathers |
| 17. k47r - Original rank: 16. Authored by Richard Huford | 38. k68r - Original rank: 12. Authored by Francois Leyvraz |
| 18. k48r - Original rank: 53. Authored by George Hufford | 39. k69r - Original rank: 29. Authored by Johann Joss |
| 19. k49r - Original rank: 4. Authored by Rob Cave | 40. k70r - Original rank: 32. Authored by Robert Pebly |
| 20. k50r - Original rank: 54. Authored by Rik Smoody | 41. k71r - Original rank: 60. Authored by James E Hall |
| 21. k51r - Original rank: 18. Authored by John William Colbert | |

- | | |
|---|--|
| 42. k72r - Original rank: 13. Authored by Edward C White Jr | 53. k83r - Original rank: 15. Authored by Paul E Black |
| 43. k73r - Original rank: 41. Authored by George Zimmerman | 54. k84r - Original rank: 9. Authored by T Nicolaus Tideman and Paula Chieruzz |
| 44. k74r - Original rank: 61. Authored by Edward Friedland | 55. k85r - Original rank: 33. Authored by Robert B Falk and James M Langsted |
| 45. k75r - Original rank: 8. Authored by Paul D Harrington | 56. k86r - Original rank: 28. Authored by Bernard Grofman |
| 46. k76r - Original rank: 46. Authored by David Gladstein | 57. k87r - Original rank: 44. Authored by E E H Schurmann |
| 47. k77r - Original rank: 55. Authored by Scott Feld | 58. k88r - Original rank: 22. Authored by Scott Appold |
| 48. k78r - Original rank: 19. Authored by Fred Mauk | 59. k89r - Original rank: 56. Authored by Gene Snodgrass |
| 49. k79r - Original rank: 26. Authored by Dennis Ambuehl and Kevin Hickey | 60. k90r - Original rank: 24. Authored by John Maynard Smith |
| 50. k80r - Original rank: 36. Authored by Robyn M Dawes and Mark Batell | 61. k91r - Original rank: 30. Authored by Jonathan Pinkley |
| 51. k81r - Original rank: 43. Authored by Martyn Jones | 62. k92r - Original rank: 1. Authored by Anatol Rapoport |
| 52. k82r - Original rank: 49. Authored by Robert A Leyland | 63. krandmc - Original rank: 62. Authored by None |

S2 Original submissions and their tournament rankings

Supplementary Table 1 lists the top 15 strategies from Axelrod's original tournament along with their rankings in our reproduced tournament.

For the original strategies, we evaluate performance in the following tournaments:

- **Reproduced tournament:** A faithful reproduction of Axelrod's second tournament.
- **Noisy tournament:** The original tournament with a noise (error) probability of 0.01.
- **Stewart and Plotkin tournament [2]:** A tournament with the original submission and the submissions used by Stewart & Plotkin.
- **Axelrod-Python tournament:** A tournament including all strategies from the **Axelrod-Python** library and the original submissions.
- **Axelrod-Python tournament with 1% noise:** As above, with 1% noise.
- **Axelrod-Python tournament with 5% noise:** As above, with 5% noise.

Supplementary Table 2 shows the strategies and their ranks across these tournaments.

	Author	Average Score	Reproduced Rank	Original Rank	Average Cooperation Rate	Win proportion
k92r	Anatol Rapoport	2.8785	1	1	0.9218	0.6306
k61r	Danny C Champion	2.7907	12	2	0.9542	0.0000
k42r	Otto Borufsen	2.8607	2	3	0.9161	0.3149
k49r	Rob Cave	2.8260	4	4	0.8922	0.0000
k44r	William Adams	2.8260	3	5	0.8819	0.0038
k60r	Jim Graaskamp and Ken Katzen	2.8102	9	6	0.8440	0.0102
k41r	Herb Weiner	2.8107	8	7	0.8650	0.0000
k75r	Paul D Harrington	2.8258	5	8	0.8019	0.0224
k84r	T Nicolaus Tideman and Paula Chieruzz	2.8124	7	9	0.8818	0.0000
k32r	Charles Kluepfel	2.8170	6	10	0.8726	0.0080
k35r	Abraham Getzler	2.8010	11	11	0.8878	0.0002
k68r	Fransois Leyvraz	2.8085	10	12	0.9169	0.0000
k72r	Edward C White Jr	2.7762	13	13	0.9242	0.0000
k46r	Graham J Eatherley	2.7722	14	14	0.9483	0.0000
k83r	Paul E Black	2.7425	15	15	0.9347	0.0000

Supplementary Table 1: **Top 15 strategies in the reproduced tournament.** The table lists the top 15 strategies from Axelrod’s original tournament along with their rankings in our reproduced tournament. For each strategy, we report the average score and the average cooperation rate in the reproduced tournament, as well as the proportion of wins across our repetitions of the original tournament. *TFT* wins 63.3% of these repetitions, indicating that in 36.7% of runs a different winner emerges.

S3 Reproducing Champion and Axelrod’s linear model

Here we provide additional detail on the reproduction of Axelrod’s second tournament. Supplementary Figure 1 shows the corrected Fortran source code of *Champion* (k61r), and Supplementary Figure 2 reproduces Axelrod’s linear model relating strategy features to average score.

```

FUNCTION K61R(ISPICK,ITURN,K,L,R, JA)
C BY DANNY C. CHAMPION
C TYPED BY JM 3/27/79
~~~~~
k61r=ja ! Added 7/27/93 to report own old value
IF (ITURN .EQ. 1) ICOOP = 0 ! Added 10/8/2017 to fix bug for multiple runs
IF (ITURN .EQ. 1) K61R = 0
IF (ISPICK .EQ. 0) ICOOP = ICOOP + 1
IF (ITURN .LE. 10) RETURN
K61R = ISPICK
IF (ITURN .LE. 25) RETURN
K61R = 0
COPRAT = FLOAT(ICOOP) / FLOAT(ITURN)
IF (ISPICK .EQ. 1 .AND. COPRAT .LT. .6 .AND. R .GT. COPRAT)
+K61R = 1
RETURN
END
%DIFF >

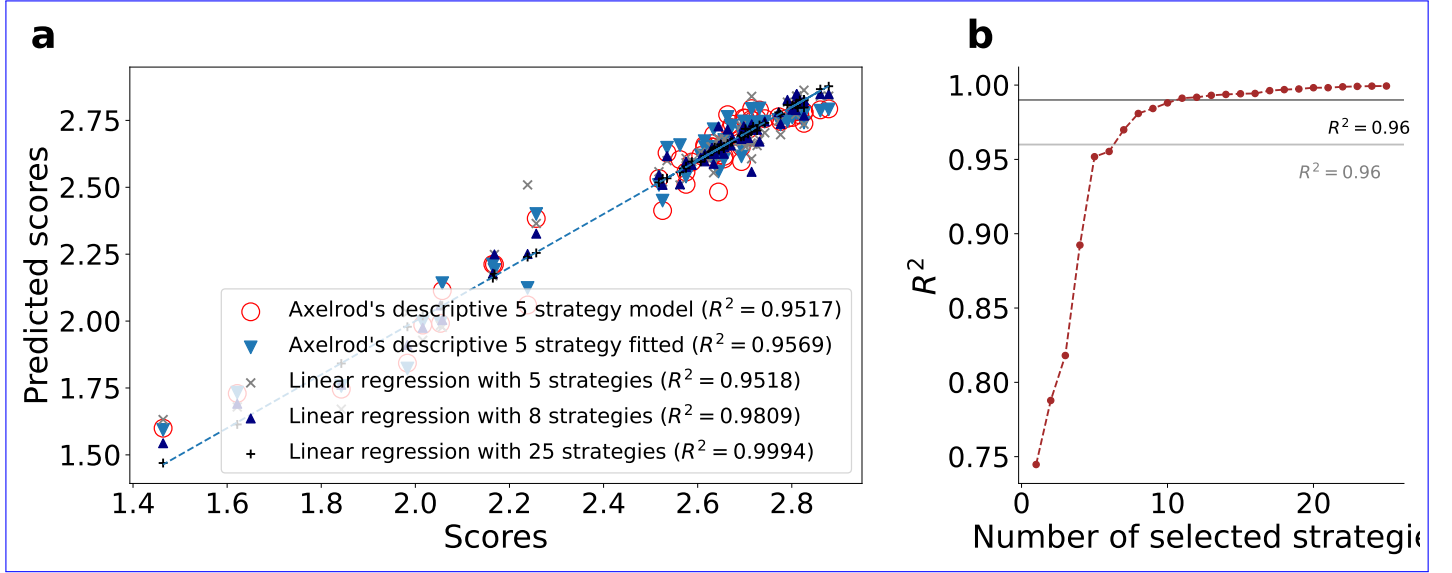
```

Supplementary Figure 1: **Fortran source code for k61r (Champion).** The original implementation contained a bug: the variable *ICOOP* was not initialized within the function. In the first match it effectively started at 0, but in subsequent matches it could have kept its value from the previous match or be reset to 0, depending on the compiler and runtime environment. Line 5 shows the fix, which explicitly initializes *ICOOP* at the start of each match.

S4 Full list of strategies and ranks of Stewart and Plotkin tournament

	Original Rank	Reproduced Rank	Reproduced Rank (Noise 1%)	S & P tournament Rank	AXL tournament Rank	AXL (Noise 1%) tournament Rank	AXL (Noise 5%) tournament Rank
k92r	1	1	28	4	57	70	69
k61r	2	12	44	8	85	126	156
k42r	3	2	8	2	16	25	21
k49r	4	4	12	5	30	32	23
k44r	5	3	16	7	46	37	26
k60r	6	9	11	16	19	31	25
k41r	7	8	13	10	35	33	40
k75r	8	5	20	6	110	42	31
k84r	9	7	21	15	47	43	76
k32r	10	6	18	12	23	40	34
k35r	11	11	36	14	87	95	89
k68r	12	10	41	9	66	117	139
k72r	13	13	32	13	78	77	38
k46r	14	14	26	11	74	63	77
k83r	15	15	35	17	106	92	83
k47r	16	16	4	22	84	16	5
k64r	17	17	29	18	92	71	46
k51r	18	19	54	30	164	167	158
k78r	19	20	33	37	145	80	61
k66r	20	18	19	19	71	41	19
k58r	21	21	17	23	25	38	43
k88r	22	22	43	25	126	122	113
k31r	23	23	39	24	127	107	73
k90r	24	26	38	27	132	104	71
k39r	25	27	34	21	130	81	20
k79r	26	24	23	26	88	52	54
k67r	27	25	37	28	146	103	33
k86r	28	30	15	33	67	36	49
k69r	29	31	42	43	144	121	48
k91r	30	28	27	29	89	67	52
k57r	31	29	49	40	115	147	182
k70r	32	35	46	41	104	129	149
k85r	33	37	24	47	20	57	101
k38r	34	33	9	34	51	26	7
k40r	35	32	14	38	27	34	80
k80r	36	34	3	31	22	10	3
k37r	37	42	30	44	113	73	57
k56r	38	41	6	36	31	21	12
k43r	39	38	5	32	79	18	6
k59r	40	40	7	35	32	23	11
k73r	41	44	22	42	101	44	50
k55r	42	43	10	39	36	27	15
k81r	43	45	2	45	43	6	14
k87r	44	39	31	46	26	74	124
k53r	45	46	45	49	99	128	128
k76r	46	36	53	48	155	166	155
k65r	47	48	1	50	28	5	18
k52r	48	47	50	51	169	151	100
k82r	49	50	25	56	148	58	28
k45r	50	49	52	55	163	159	115
k62r	51	51	51	57	173	158	99
k34r	52	52	40	58	83	111	119
k48r	53	53	48	60	151	141	94
k50r	54	54	60	62	240	235	249
k77r	55	55	47	64	170	131	45
k89r	56	56	58	63	175	200	191
k63r	57	58	61	67	230	240	248
k54r	58	57	57	65	191	191	143
k33r	59	59	55	66	166	178	145
k71r	60	60	59	68	219	222	220
k74r	61	61	56	69	190	186	120
krandomc	62	62	63	71	260	252	255
k36r	63	63	62	72	247	251	243

Supplementary Table 2: **Ranks of the original strategies across different tournaments.**



Supplementary Figure 2: **Reproduction of Axelrod's linear model.** **a**, The representative strategies and their effect on average score. We reproduce the linear model described in [1], which reports $R^2 = 0.979$. Fitting a new model to the same five strategies yields $R^2 = 0.957$. The fit is an ordinary least squares linear regression; the reported R^2 values are coefficients of determination, and no significance tests or multiple-comparison adjustments were applied. **b**, R^2 as a function of the number of strategies included (subset size), obtained using recursive feature elimination (RFE) [3]. Starting from the full set of strategies, we iteratively remove the least informative one, refit the model, and record the resulting R^2 . For each subset size (from 1 to 25), the subset yielding the highest R^2 is retained. The estimated R^2 rises rapidly from about 0.75 to 0.95 as the number of included strategies increases from 1 to 5, after which improvements are smaller.

The full list of strategies and their ranks Supplementary Table 3 reports the top 15 strategies in the Stewart and Plotkin tournament [2], and Supplementary Table 4 gives the full list of strategies and their ranks in that tournament, which now includes the original strategies, is shown in Table 4.

Name	Average Score	Rank	Average Cooperation Rate	Reproduced Average Cooperation Rate	S. and P. Average Cooperation Rate
ZD-GTFT-2: 0.25, 0.5	2.834	1	0.926	-	0.855
k42r	2.825	2	0.903	0.916	-
GTFT: 0.33	2.819	3	0.941	-	0.893
k92r	2.812	4	0.888	0.922	0.731
k49r	2.787	5	0.879	0.892	-
k75r	2.786	6	0.782	0.802	-
k44r	2.781	7	0.866	0.882	-
k61r	2.773	8	0.945	0.954	-
k68r	2.768	9	0.904	0.917	-
k41r	2.761	10	0.847	0.865	-
k46r	2.758	11	0.939	0.948	-
k32r	2.758	12	0.849	0.873	-
k72r	2.756	13	0.915	0.924	-
k35r	2.747	14	0.863	0.888	-
k84r	2.745	15	0.861	0.882	-

Supplementary Table 3: **Top 15 strategies in the Stewart and Plotkin tournament.** We report the mean score, rank, and mean cooperation rate. For comparison, we also include the cooperation rate from our reproduced tournament, alongside the cooperation rates originally reported by Stewart and Plotkin.

	Mean Score	Rank	Mean Cooperation Rate	Reproduced Mean Cooperation Rate	S. and P. Mean Cooperation Rate
ZD-GTFT-2: 0.25, 0.5	2.834	1	0.926	-	0.855
k42r	2.825	2	0.903	0.916	-
GTFT: 0.33	2.819	3	0.941	-	0.893
k92r	2.812	4	0.888	0.922	0.731
k49r	2.787	5	0.879	0.892	-
k75r	2.786	6	0.782	0.802	-
k44r	2.781	7	0.866	0.882	-
k61r	2.773	8	0.945	0.954	-
k68r	2.768	9	0.904	0.917	-
k41r	2.761	10	0.847	0.865	-
k46r	2.758	11	0.939	0.948	-
k32r	2.758	12	0.849	0.873	-
k72r	2.756	13	0.915	0.924	-
k35r	2.747	14	0.863	0.888	-
k84r	2.745	15	0.861	0.882	-
k60r	2.735	16	0.814	0.844	-
k83r	2.725	17	0.922	0.935	-
k64r	2.710	18	0.876	0.884	-
k66r	2.702	19	0.861	0.866	-
Hard Tit For 2 Tats	2.701	20	0.926	-	0.807
k39r	2.700	21	0.671	0.676	-
k47r	2.696	22	0.709	0.723	-
k58r	2.694	23	0.846	0.852	-
k31r	2.690	24	0.908	0.920	-
k88r	2.684	25	0.851	0.861	-
k79r	2.678	26	0.864	0.875	-
k90r	2.676	27	0.943	0.964	0.844
k67r	2.676	28	0.743	0.749	-
k91r	2.662	29	0.889	0.873	-
k51r	2.646	30	0.892	0.905	-
k80r	2.641	31	0.813	0.809	-
k43r	2.641	32	0.673	0.666	-
k86r	2.639	33	0.803	0.811	-
k38r	2.639	34	0.865	0.874	-
k59r	2.635	35	0.889	0.891	-
k56r	2.635	36	0.890	0.891	-
k78r	2.634	37	0.720	0.755	-
k40r	2.625	38	0.742	0.746	-
k55r	2.620	39	0.903	0.903	-
k57r	2.620	40	0.799	0.811	-
k70r	2.596	41	0.752	0.769	-
k73r	2.595	42	0.761	0.764	-
k69r	2.591	43	0.672	0.702	-
k37r	2.590	44	0.778	0.791	-
k81r	2.589	45	0.774	0.779	-
k87r	2.586	46	0.760	0.766	-
k85r	2.584	47	0.713	0.726	-
k76r	2.576	48	0.636	0.680	-
k53r	2.568	49	0.730	0.728	-
k65r	2.559	50	0.718	0.715	-
k52r	2.545	51	0.820	0.838	-
Win-Stay Lose-Shift	2.544	52	0.809	-	0.808
Hard Tit For Tat	2.539	53	0.710	-	0.663
Cooperator	2.522	54	1.000	-	1.000
k45r	2.521	55	0.726	0.754	-
k82r	2.491	56	0.529	0.544	-
k62r	2.490	57	0.732	0.754	-
k34r	2.478	58	0.647	0.652	0.655
Hard Go By Majority	2.375	59	0.658	-	0.459
k48r	2.300	60	0.504	0.499	-
First by Joss: 0.9	2.280	61	0.501	-	0.345
k50r	2.248	62	0.929	0.930	-
k89r	2.212	63	0.504	0.498	-
k77r	2.165	64	0.327	0.332	-
k54r	2.118	65	0.483	0.467	-
k33r	2.108	66	0.424	0.411	-
k63r	2.096	67	0.502	0.502	-
k71r	2.032	68	0.166	0.158	-
k74r	1.881	69	0.171	0.174	-
ZD-Extort-2: 0.1111111111111111, 0.5	1.838	70	0.254	-	0.232
krandomc	1.698	71	0.500	0.500	0.500
k36r	1.520	72	0.092	0.092	-
Defector	1.472	73	0.000	-	0.000

Supplementary Table 4: Stewart and Plotkin tournament full list of rankings.

S5 Full list of strategies and ranks for Axelrod-Python tournament

The full list of strategies and their ranks in the Axelrod-Python tournament, which now includes the original strategies, can be found online at <https://github.com/Axelrod-Python/tournament>. This includes tournaments with and without noise. [Supplementary Table 5 reports the top 16 strategies in this tournament.](#)

	Average Score	Rank	Average Cooperation Rate	Original Rank	Reproduced Average Cooperation Rate	Library Rank	Library Average Cooperation Rate
EvolvedLookerUp2_2_2 [†]	2.802	1	0.756	-	-	1	0.736
Evolved HMM 5 [†]	2.788	2	0.751	-	-	2	0.742
Omega TFT: 3, 8	2.786	3	0.750	-	-	12	0.725
Evolved ANN 5 [†]	2.784	4	0.720	-	-	3	0.713
Evolved ANN [†]	2.782	5	0.739	-	-	5	0.727
Evolved FSM 16 [†]	2.780	6	0.731	-	-	4	0.713
Evolved FSM 16 Noise 05 [†]	2.776	7	0.726	-	-	6	0.717
PSO Gambler 2_2_2 [†]	2.760	8	0.702	-	-	7	0.692
Original Gradual	2.757	9	0.808	-	-	-	-
PSO Gambler 2_2_2 Noise 05 [†]	2.756	10	0.740	-	-	14	0.723
PSO Gambler Mem1 [†]	2.756	11	0.734	-	-	9	0.728
Evolved FSM 4 [†]	2.752	12	0.793	-	-	10	0.777
PSO Gambler 1_1_1 [†]	2.752	13	0.704	-	-	8	0.702
Gradual	2.746	14	0.763	-	-	15	0.793
DBS: 0.75, 3, 4, 3, 5	2.739	15	0.750	-	-	11	0.739
k42r	2.735	16	0.820	3	0.916	-	-

Supplementary Table 5: [Top 16 strategies in the Axelrod-Python tournament.](#) This tournament includes all strategies from the Axelrod-Python library alongside the original Fortran strategies. In total, there are 272 strategies competing. Strategies denoted by [†] were not designed but, they were trained via reinforcement learning, as described in [4].

S6 The expanded Axelrod-Python tournament under noise

We repeat the expanded Axelrod-Python tournament under noise, where with a fixed probability each intended action is flipped. [Supplementary Figure 3 shows the results under 1% noise and Supplementary Figure 4 under 5% noise.](#) As expected, noise reduces the advantage of highly cooperative strategies, but the overall structure of the tournament is preserved.

S7 Fortran source code for k42r

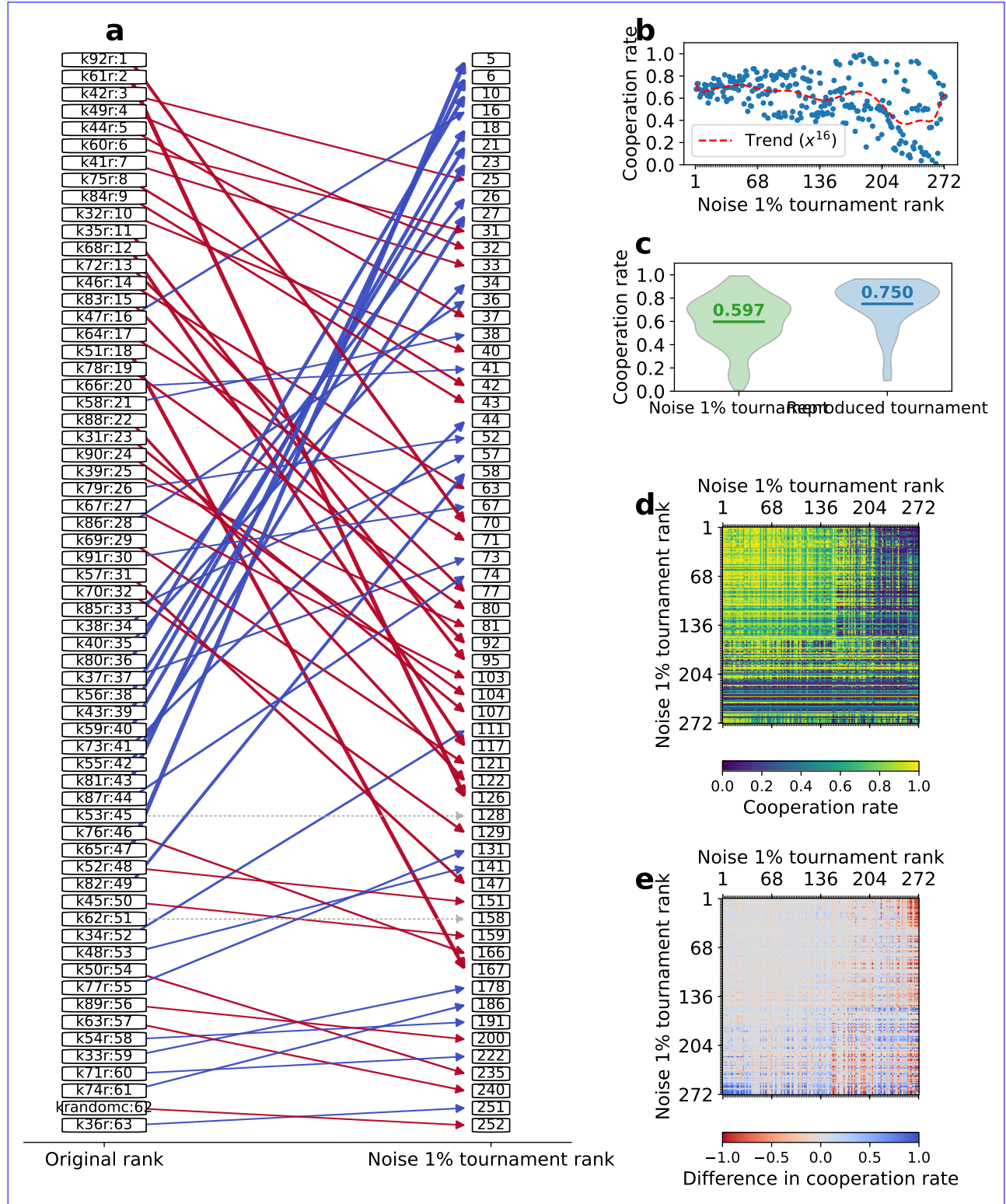
From inspection of the source code, the behaviour of the strategy can be described as follows (these descriptions are taken from the re-implementation in Python from [5], which has been verified to reproduce the original Fortran behaviour accurately):

This player keeps track of the opponent's responses to own behavior:

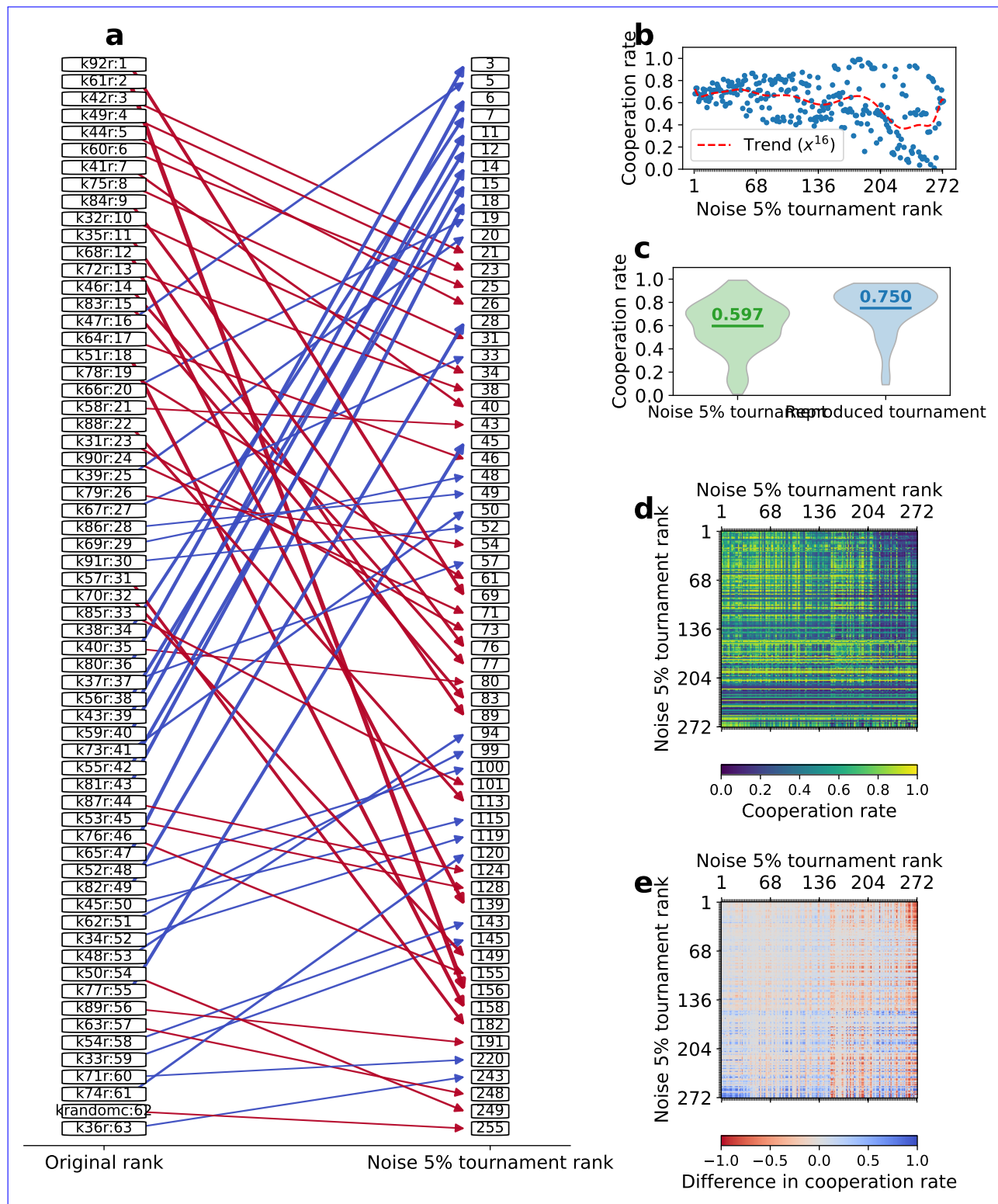
- `cd_count` counts: Opponent cooperates as response to player defecting.
- `cc_count` counts: Opponent cooperates as response to player cooperating.

The player has a defect mode and a normal mode. In defect mode, the player will always defect. In normal mode, the player obeys the following ranked rules:

1. If in the last three turns, both the player/opponent defected, then cooperate for a single turn.
2. If in the last three turns, the player/opponent acted differently from each other and they're alternating, then change next defect to cooperate. (Doesn't block third rule.)



Supplementary Figure 3: [Reproduction of Axelrod's second tournament with the addition of the strategies from the Axelrod-Python library and 1% noise.](#) **a**, Comparison of the original rankings (left column) and the rankings in this tournament (right column), with strategies ordered by their original ranks. Arrows indicate the change in rank, with widths proportional to the size of the change; blue arrows indicate a positive change in rank, red arrows a negative change, and grey arrows no change. **b**, Cooperation rates of strategies ordered by rank in this tournament. **c**, Tournament cooperation rates across all repetitions for this tournament (green) and the reproduced tournament without the additional strategies (blue); the solid line shows the average cooperation rate. **d**, Pairwise cooperation rates between strategies, ordered by their rank in this tournament. **e**, Differences in pairwise cooperation rates, again ordered by rank in this tournament.



Supplementary Figure 4: [Reproduction of Axelrod's second tournament with the addition of the strategies from the Axelrod-Python library and 5% noise](#). The panels follow the same layout as Supplementary Figure 3. **a**, Comparison of the original rankings (left column) and the rankings in this tournament (right column), with strategies ordered by their original ranks. Arrows indicate the change in rank, with widths proportional to the size of the change; blue arrows indicate a positive change in rank, red arrows a negative change, and grey arrows no change. **b**, Cooperation rates of strategies ordered by rank in this tournament. **c**, Tournament cooperation rates across all repetitions for this tournament (green) and the reproduced tournament without the additional strategies (blue); the solid line shows the average cooperation rate. **d**, Pairwise cooperation rates between strategies, ordered by their rank in this tournament. **e**, Differences in pairwise cooperation rates, again ordered by rank in this tournament.

3. Otherwise, do tit-for-tat.

Start in normal mode, but every 25 turns starting with the 27th turn, re-evaluate the mode. Enter defect mode if any of the following conditions hold:

- Detected random: Opponent cooperated 7-18 times since last mode evaluation (or start) AND less than 70% of opponent cooperation was in response to player's cooperation, i.e. $cc_count / (cc_count + cd_count) < 0.7$
- Detect defective: Opponent cooperated fewer than 3 times since last mode evaluation.

When switching to defect mode, defect immediately. The first two rules for normal mode require that the last three turns were in normal mode. When starting normal mode from defect mode, defect on first move.

S8 List of all strategies

Here we list all the strategies including both the original and the ones from the `Axelrod-Python` library. Their ranks are shown as: Rank in full tournament / Rank in full tournament with 1% noise / Rank in full tournament with 5% noise / Rank in original tournament (if an original strategy).

- | | |
|---|--|
| 1. EvolvedLookerUp2_2_2: 1 / 30 / 118 | 19. k60r: 19 / 31 / 25 / 6 |
| 2. Evolved HMM 5: 2 / 13 / 59 | 20. k85r: 20 / 57 / 101 / 33 |
| 3. Omega TFT: 3, 8: 3 / 4 / 17 | 21. EugeneNier: (D,): 21 / 35 / 27 |
| 4. Evolved ANN 5: 4 / 7 / 36 | 22. k80r: 22 / 10 / 3 / 36 |
| 5. Evolved ANN: 5 / 3 / 8 | 23. k32r: 23 / 40 / 34 / 10 |
| 6. Evolved FSM 16: 6 / 15 / 78 | 24. DoubleCrosser: (D, D): 24 / 11 / 13 |
| 7. Evolved FSM 16 Noise 05: 7 / 1 / 4 | 25. k58r: 25 / 38 / 43 / 21 |
| 8. PSO Gambler 2_2_2: 8 / 53 / 138 | 26. k87r: 26 / 74 / 124 / 44 |
| 9. Original Gradual: 9 / 20 / 30 | 27. k40r: 27 / 34 / 80 / 35 |
| 10. PSO Gambler 2_2_2 Noise 05: 10 / 9 / 24 | 28. k65r: 28 / 5 / 18 / 47 |
| 11. PSO Gambler Mem1: 11 / 39 / 107 | 29. Forgetful Fool Me Once: 0.05: 29 / 19 / 41 |
| 12. Evolved FSM 4: 12 / 28 / 85 | 30. k49r: 30 / 32 / 23 / 4 |
| 13. PSO Gambler 1_1_1: 13 / 61 / 140 | 31. k56r: 31 / 21 / 12 / 38 |
| 14. Gradual: 14 / 17 / 29 | 32. k59r: 32 / 23 / 11 / 40 |
| 15. DBS: 0.75, 3, 4, 3, 5: 15 / 2 / 1 | 33. Revised Downing: 33 / 22 / 10 |
| 16. k42r: 16 / 25 / 21 / 3 | 34. Fool Me Once: 34 / 14 / 37 |
| 17. Winner12: 17 / 60 / 127 | 35. k41r: 35 / 33 / 40 / 7 |
| 18. Spiteful Tit For Tat: 18 / 24 / 22 | 36. k55r: 36 / 27 / 15 / 42 |
| | 37. First by Shubik: 37 / 87 / 95 |

```

      FUNCTION K42R(JPICK,MOVEN,ISCORE,JSCORE,RANDOM,JA)
C BY OTTO BORUFSEN
C TYPED FROM FORTRAN BY AX, 1/25/79
      DIMENSION MHIST(2,2)
      K42R=JA      ! Added 7/27/93 to report own old value
C INITIALIZE FIRST MOVE
      IF(MOVEN.NE.1)GOTO 20
      L3MOV=0
      L3ECH=0
      IDEF=0
      ICOOP=0
      IPICK=0
      DO 10 I=1,2
      DO 10 J=1,2
10      MHIST(I,J)=0
      GO TO 500
20      IF(MOVEN.EQ.2)GOTO 25
C UPDATE MOVE HISTORY
      MHIST(I2PCK+1,JPICK+1)=MHIST(I2PCK+1,JPICK+1)+1
25      IF(IDEF.EQ.0)GOTO 30
C OPPONENT HAS BEEN PROVED "RANDOM" OR
C "DEFECTIVE",I DEFECT FOR 25 MOVES
      K42R=1
      GO TO 100
30      IF(IPICK.EQ.0.OR.JPICK.EQ.0)GOTO 40
C MUTUAL DEFECTIONS ON LAST MOVE.
      L3MOV=L3MOV+1
      IF(L3MOV.LT.3)GOTO 50
C MUTUAL DEFECTION ON
C LAST THREE MOVES.
C I COOPERATE ONCE ON NEXT MOVE.
      K42R=0
      L3MOV=0
      L3ECH=0
      GO TO 100
C ONE (OR BOTH) COOPERATED ON LAST MOVE.
40      L3MOV=0
      IF(IPICK.EQ.JPICK)GOTO 45
      IF(JPICK.NE.I2PCK.OR.IPICK.NE.J2PCK)GOTO 45
C ECHO-EFFECT ON LAST MOVE.
      L3ECH=L3ECH+1
      IF(L3ECH.LT.3)GOTO 50
C ECHO-EFFECT ON LAST THREE MOVES.
C MY NEXT DEFECTION WILL BE SUBSTITUTED
C BY A COOPERATION.
      L3ECH=0
      L3MOV=0
      ICOOP=1
      GO TO 50
45      L3ECH=0
C PLAY 'TIT FOR TAT' AS MAIN RULE.
50      K42R=JPICK
100     IF(MOD(MOVEN-2,25).NE.0.OR.MOVEN.EQ.2)GOTO 650
C ON EVERY 25 MOVES:
C CHECK IF OPPONENT SEEMS TO BE
C 'RANDOM' OR 'DEFECTIVE'.
      IDEF=0
      JNCOP=MHIST(1,1)+MHIST(2,1)
C IS OPPONENT 'RANDOM'?
      IF(JNCOP.GT.17)GOTO 155
      IF(JNCOP.LT.8)GOTO 130
      IF(100*MHIST(1,1)/JNCOP.LT.70)IDEF=1
      GO TO 155
C IS OPPONENT 'DEFECTIVE'?
130     IF(JNCOP.LT.3)IDEF=1
155     DO 160 I=1,2
      DO 160 J=1,2
160     MHIST(I,J)=0
      IF(IDEF.EQ.0)GOTO 650
C OPPONENT SEEMS TO BE
C 'RANDOM' OR 'DEFECTIVE'.
C I DEFECT FOR NEXT 25 MOVES.
      ICOOP=0
      L3MOV=0
      L3ECH=0
      GO TO 600
C I COOPERATE.
500     K42R=0
      GO TO 650
C I DEFECT.
600     K42R=1
650     IF(ICOOP.EQ.0.OR.K42R.EQ.0)GOTO 660
      ICOOP=0
      K42R=0
660     I2PCK=IPICK
      J2PCK=JPICK
      IPICK=K42R
      RETURN
      END

```

Supplementary Figure 5: Fortran Source code for original k42r strategy submitted to Axelrod's second tournament.

38. ZD-GTFT-2: 0.25, 0.5: 38 / 48 / 63
39. Soft Joss: 0.9: 39 / 49 / 56
40. BackStabber: (D, D): 40 / 8 / 9
41. MEM2: 41 / 79 / 103
42. AON2: 42 / 116 / 168
43. k81r: 43 / 6 / 14 / 43
44. Meta Hunter: 6 players: 44 / 69 / 108
45. NMWE Finite Memory: 79 players: 45 / 91 / 126
46. k44r: 46 / 37 / 26 / 5
47. k84r: 47 / 43 / 76 / 9
48. GTFT: 0.33: 48 / 55 / 72
49. EvolvedLookerUp1_1_1: 49 / 115 / 147
50. NMWE Stochastic: 67 players: 50 / 97 / 123
51. k38r: 51 / 26 / 7 / 34
52. Michaelos: (D,): 52 / 86 / 97
53. First by Stein and Rapoport: 0.05: (D, D): 53 / 47 / 44
54. NMWE Memory One: 36 players: 54 / 90 / 104
55. Resurrection: 55 / 65 / 67
56. Nice Meta Winner Ensemble: 213 players: 56 / 98 / 111
57. k92r: 57 / 70 / 69 / 1
58. Adaptive Tit For Tat: 0.5: 58 / 72 / 68
59. Forgiving Tit For Tat: 59 / 46 / 58
60. NMWE Deterministic: 146 players: 60 / 93 / 109
61. General Soft Grudger: n=1,d=4,c=2: 61 / 110 / 144
62. Soft Grudger: 62 / 109 / 142
63. Meta Majority Long Memory: 134 players: 63 / 62 / 62
64. Firm But Fair: 64 / 100 / 91
65. Level Punisher: 65 / 29 / 16
66. k68r: 66 / 117 / 139 / 12
67. k86r: 67 / 36 / 49 / 28
68. NMWE Long Memory: 134 players: 68 / 105 / 112
69. Meta Majority: 213 players: 69 / 66 / 75
70. ZD-GEN-2: 0.125, 0.5, 3: 70 / 99 / 110
71. k66r: 71 / 41 / 19 / 20
72. Evolved ANN 5 Noise 05: 72 / 12 / 2
73. Second by Eatherley: 73 / 64 / 79
74. k46r: 74 / 63 / 77 / 14
75. TF3: 75 / 75 / 90
76. Memory Decay: 0.1, 0.03, -2, 1, Tit For Tat, 15: 76 / 51 / 51
77. Nice Meta Winner: 213 players: 77 / 114 / 129
78. k72r: 78 / 77 / 38 / 13
79. k43r: 79 / 18 / 6 / 39
80. Doubler: 80 / 54 / 42
81. Meta Majority Finite Memory: 79 players: 81 / 59 / 64
82. GrudgerAlternator: 82 / 124 / 159
83. k34r: 83 / 111 / 119 / 52
84. k47r: 84 / 16 / 5 / 16
85. k61r: 85 / 126 / 156 / 2
86. Second by Champion: 86 / 108 / 93
87. k35r: 87 / 95 / 89 / 11
88. k79r: 88 / 52 / 54 / 26
89. k91r: 89 / 67 / 52 / 30
90. ShortMem: 90 / 56 / 32
91. Dynamic Two Tits For Tat: 91 / 94 / 92
92. k64r: 92 / 71 / 46 / 17
93. Adaptive Pavlov 2011: 93 / 112 / 121

94. First by Tideman and Chieruzzi: (D, D): 94 / 120 / 102
95. Adaptive Pavlov 2006: 95 / 113 / 131
96. Meta Majority Memory One: 36 players: 96 / 101 / 87
97. Once Bitten: 97 / 45 / 39
98. First by Davis: 10: 98 / 127 / 122
99. k53r: 99 / 128 / 128 / 45
100. Soft Go By Majority: 5: 100 / 50 / 35
101. k73r: 101 / 44 / 50 / 41
102. Limited Retaliate 3: 0.05, 20: 102 / 130 / 152
103. Inverse Punisher: 103 / 132 / 157
104. k70r: 104 / 129 / 149 / 32
105. Forgiver: 105 / 82 / 96
106. k83r: 106 / 92 / 83 / 15
107. Limited Retaliate 2: 0.08, 15: 107 / 134 / 164
108. Slow Tit For Two Tats 2: 108 / 68 / 74
109. First by Grofman: 109 / 135 / 176
110. k75r: 110 / 42 / 31 / 8
111. Stalker: (D,): 111 / 88 / 65
112. VeryBad: 112 / 89 / 60
113. k37r: 113 / 73 / 57 / 37
114. Hard Tit For 2 Tats: 114 / 83 / 70
115. k57r: 115 / 147 / 182 / 31
116. Soft Go By Majority: 10: 116 / 76 / 47
117. Soft Go By Majority: 20: 117 / 84 / 55
118. Soft Go By Majority: 40: 118 / 96 / 66
119. Limited Retaliate: 0.1, 20: 119 / 138 / 170
120. Nice Average Copier: 120 / 123 / 133
121. Worse and Worse 3: 121 / 125 / 132
122. Punisher: 122 / 137 / 167
123. Retaliate 2: 0.08: 123 / 139 / 163
124. Second by Appold: 124 / 119 / 106
125. Retaliate 3: 0.05: 125 / 133 / 153
126. k88r: 126 / 122 / 113 / 22
127. k31r: 127 / 107 / 73 / 23
128. Math Constant Hunter: 128 / 136 / 146
129. Forgetful Grudger: 129 / 144 / 172
130. k39r: 130 / 81 / 20 / 25
131. Retaliate: 0.1: 131 / 143 / 171
132. k90r: 132 / 104 / 71 / 24
133. Grumpy: Nice, 10, -10: 133 / 118 / 88
134. N Tit(s) For M Tat(s): 3, 2: 134 / 85 / 81
135. Thumper: 135 / 148 / 161
136. Two Tits For Tat: 136 / 146 / 165
137. Inverse: 137 / 140 / 148
138. Delayed AON1: 138 / 152 / 215
139. Hard Tit For Tat: 139 / 150 / 178
140. Tricky Level Punisher: 140 / 106 / 117
141. Appeaser: 141 / 155 / 205
142. Win-Stay Lose-Shift: 142 / 156 / 206
143. AdaptorBrief: 143 / 78 / 53
144. k69r: 144 / 121 / 48 / 29
145. k78r: 145 / 80 / 61 / 19
146. k67r: 146 / 103 / 33 / 27
147. Eventual Cycle Hunter: 147 / 157 / 198
148. k82r: 148 / 58 / 28 / 49
149. First by Nydegger: 149 / 153 / 173
150. First by Graaskamp: 0.05: 150 / 102 / 82
151. k48r: 151 / 141 / 94 / 53
152. Random Hunter: 152 / 154 / 154

153. UsuallyCooperates: 153 / 183 / 230
154. Cycle Hunter: 154 / 169 / 193
155. k76r: 155 / 166 / 155 / 46
156. Second by Tester: 156 / 163 / 151
157. Defector Hunter: 157 / 173 / 204
158. Raider: 158 / 161 / 134
159. Alternator Hunter: 159 / 175 / 200
160. AdaptorLong: 160 / 149 / 169
161. Ripoff: 161 / 172 / 162
162. Average Copier: 162 / 171 / 177
163. k45r: 163 / 159 / 115 / 50
164. k51r: 164 / 167 / 158 / 18
165. Prober 2: 165 / 162 / 130
166. k33r: 166 / 178 / 145 / 59
167. Knowledgeable Worse and Worse: 167 / 181 / 214
168. Cooperator: 168 / 179 / 211
169. k52r: 169 / 151 / 100 / 48
170. k77r: 170 / 131 / 45 / 55
171. Hard Prober: 171 / 164 / 125
172. Sneaky Tit For Tat: 172 / 180 / 179
173. k62r: 173 / 158 / 99 / 51
174. DoubleResurrection: 174 / 165 / 105
175. k89r: 175 / 200 / 191 / 56
176. Suspicious Tit For Tat: 176 / 185 / 160
177. Willing: 177 / 192 / 203
178. Remorseful Prober: 0.1: 178 / 168 / 114
179. First by Feld: 1.0, 0.5, 200: 179 / 145 / 84
180. First by Downing: 180 / 174 / 86
181. ZD-Mem2: 181 / 170 / 141
182. Worse and Worse 2: 182 / 142 / 98
183. Risky QLearner: 183 / 195 / 217
184. Cautious QLearner: 184 / 194 / 218
185. Arrogant QLearner: 185 / 196 / 216
186. Hesitant QLearner: 186 / 197 / 219
187. Detective: 187 / 189 / 166
188. Random Tit for Tat: 0.5: 188 / 188 / 180
189. First by Tullock: 189 / 160 / 116
190. k74r: 190 / 186 / 120 / 61
191. k54r: 191 / 191 / 143 / 58
192. Adaptive: 192 / 184 / 150
193. Worse and Worse: 193 / 190 / 212
194. Prober: 194 / 193 / 174
195. Hard Go By Majority: 5: 195 / 198 / 181
196. Hard Go By Majority: 196 / 202 / 194
197. Opposite Grudger: 197 / 208 / 236
198. Stochastic Cooperator: 198 / 187 / 187
199. Pun1: 199 / 206 / 199
200. Naive Prober: 0.1: 200 / 177 / 137
201. EasyGo: 201 / 220 / 259
202. First by Joss: 0.9: 202 / 176 / 136
203. ALLCorALLD: 203 / 211 / 228
204. Hard Go By Majority: 40: 204 / 201 / 188
205. Prober 4: 205 / 203 / 183
206. Prober 3: 206 / 199 / 175
207. Hard Go By Majority: 20: 207 / 207 / 185
208. Calculator: 208 / 182 / 135
209. Hard Go By Majority: 10: 209 / 209 / 186
210. SolutionB5: 210 / 210 / 184
211. Winner21: 211 / 221 / 225
212. Meta Mixer: 213 players: 212 / 205 / 195

213. Fortress3: 213 / 223 / 226
214. Fortress4: 214 / 224 / 222
215. Meta Hunter Aggressive: 7 players: 215 / 212 / 202
216. Cyclor CCCCCD: 216 / 219 / 238
217. Stochastic WSLs: 0.05: 217 / 204 / 229
218. Cyclor CCCDCD: 218 / 227 / 244
219. k71r: 219 / 222 / 220 / 60
220. Cyclor CCCD: 220 / 228 / 245
221. ZD-SET-2: 0.25, 0.0, 2: 221 / 215 / 210
222. TF2: 222 / 229 / 231
223. Cooperator Hunter: 223 / 241 / 253
224. SelfSteem: 224 / 230 / 234
225. SolutionB1: 225 / 243 / 258
226. Cyclor CCD: 226 / 231 / 239
227. Meta Winner Long Memory: 134 players: 227 / 216 / 189
228. Meta Winner: 213 players: 228 / 217 / 192
229. Alternator: 229 / 236 / 241
230. k63r: 230 / 240 / 248 / 57
231. Meta Winner Deterministic: 146 players: 231 / 218 / 190
232. ZD-Extort-2 v2: 0.125, 0.5, 1: 232 / 214 / 196
233. ZD-Extort-2: 0.1111111111111111, 0.5: 233 / 213 / 197
234. Meta Winner Ensemble: 213 players: 234 / 232 / 208
235. Meta Winner Stochastic: 67 players: 235 / 233 / 209
236. Win-Shift Lose-Stay: D: 236 / 244 / 247
237. Meta Winner Finite Memory: 79 players: 237 / 225 / 201
238. Cyclor DC: 238 / 242 / 246
239. Meta Winner Memory One: 36 players: 239 / 234 / 207
240. k50r: 240 / 235 / 249 / 54
241. TF1: 241 / 238 / 224
242. ThueMorse: 242 / 250 / 256
243. Cyclor DDC: 243 / 257 / 257
244. ZD-Extort3: 0.11538461538461539, 0.3333333333333333, 1: 244 / 226 / 213
245. AntiCyclor: 245 / 259 / 263
246. Predator: 246 / 247 / 235
247. k36r: 247 / 251 / 243 / 63
248. ZD-Extort-4: 0.23529411764705882, 0.25, 1: 248 / 239 / 221
249. ThueMorseInverse: 249 / 248 / 251
250. CollectiveStrategy: 250 / 246 / 227
251. π : 251 / 260 / 264
252. UsuallyDefects: 252 / 249 / 237
253. Tricky Cooperator: 253 / 256 / 260
254. ZD-Mischief: 0.1, 0.0, 1: 254 / 237 / 223
255. Aggravater: 255 / 254 / 233
256. Better and Better: 256 / 258 / 242
257. ZD-Extortion: 0.2, 0.1, 1: 257 / 245 / 232
258. e : 258 / 261 / 265
259. First by Anonymous: 259 / 253 / 254
260. krandomc: 260 / 252 / 255 / 62
261. Bush Mosteller: 0.5, 0.5, 3.0, 0.5: 261 / 255 / 250
262. Desperate: 262 / 263 / 262
263. Gradual Killer: (D, D, D, D, D, C, C): 263 / 266 / 267
264. Meta Minority: 213 players: 264 / 268 / 268
265. Hopeless: 265 / 265 / 266
266. Bully: 266 / 269 / 269
267. Defector: 267 / 262 / 240

268. Handshake: 268 / 264 / 252

271. Negation: 271 / 271 / 270

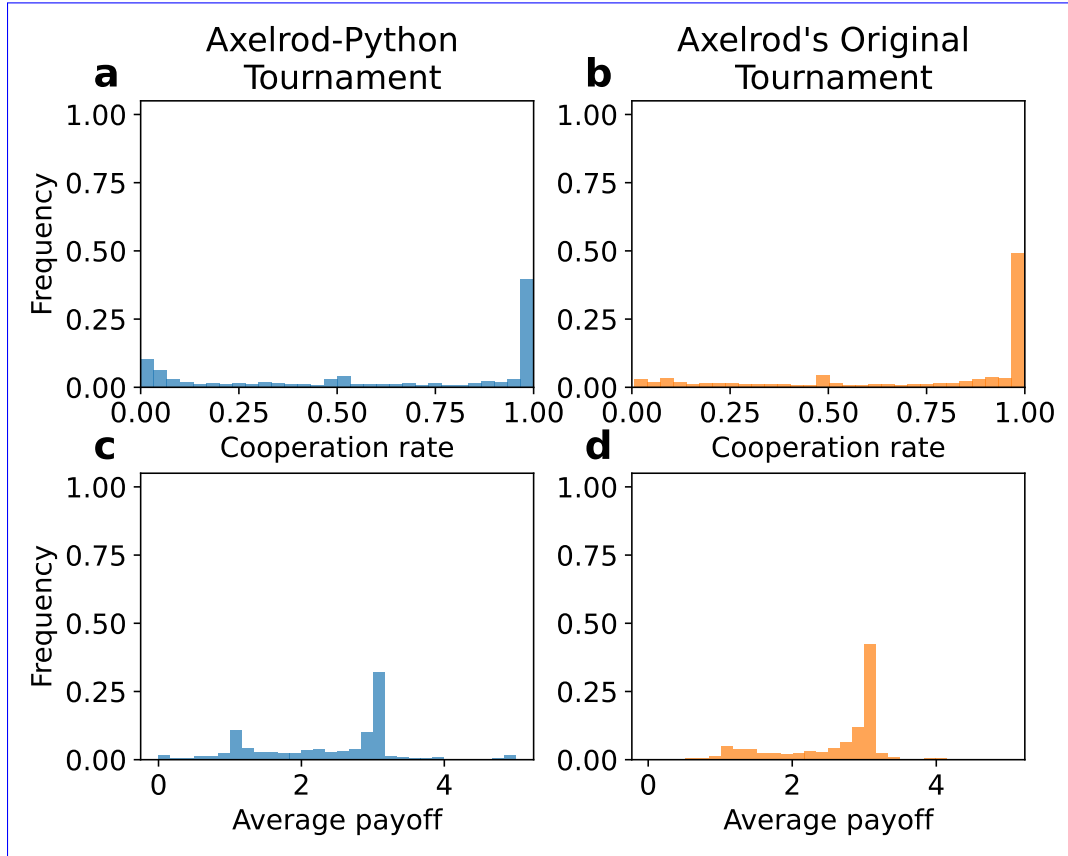
269. Tricky Defector: 269 / 267 / 261

270. ϕ : 270 / 270 / 271

272. Anti Tit For Tat: 272 / 272 / 272

S9 Interaction distributions

Here we show the distributions of interactions in terms of cooperation rates and payoffs for both the original tournament and the expanded Axelrod-Python tournament.



Supplementary Figure 6: Interaction distributions in the original and Axelrod-Python tournaments. Distributions of pairwise interaction cooperation rates and payoffs for the Axelrod-Python tournament and the original tournament. The cooperation rate distributions are strongly skewed towards one, indicating that a large fraction of interactions are highly cooperative. In the Axelrod-Python tournament, however, there is also a visible tail close to zero cooperation, indicating the presence of exploitative interactions that are largely absent from the original tournament. This difference may help explain why highly cooperative strategies such as *TFT* become less effective in more diverse tournaments, where strategies capable of exploiting others are more successful. The payoff distributions are similarly skewed around mutual cooperation payoffs, although noticeable differences between the tournaments remain. Overall, these results suggest that the interaction environments induced by the tournaments are highly structured rather than uniform across behavioural space.

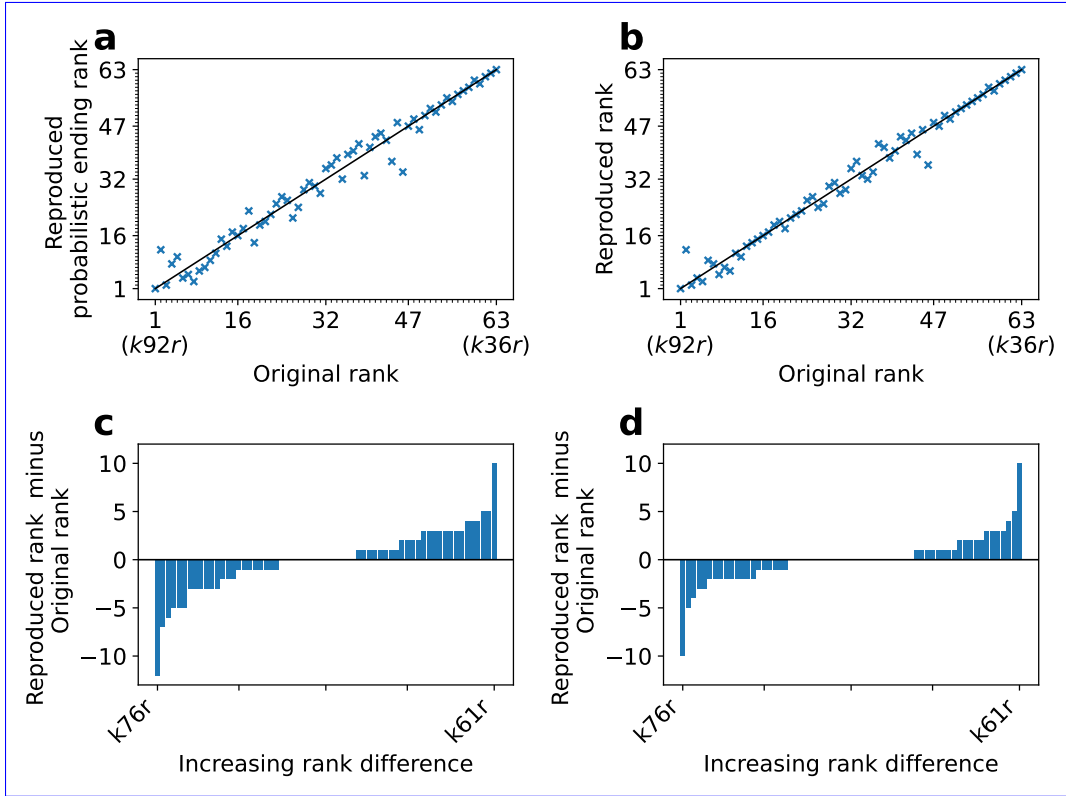
S10 Probabilistic ending tournament

Axelrod originally intended the second tournament to use probabilistic match endings rather than fixed match lengths. However, in practice, for a given continuation probability, what is described in the paper is that he sampled five different match lengths, and each pair of strategies used each of these five match lengths once. This is also the approach we followed when replicating the tournament.

Here, we perform an additional sensitivity analysis to evaluate whether our results depend sensitively on the precise interpretation of the tournament stopping rule by considering matches that end probabilistically rather than after a fixed number of turns.

Specifically, we consider a continuation probability corresponding to an expected match length of 151 turns. Under this implementation, after each turn the match continues with probability 150/151 and ends with probability 1/151. We repeat this full tournament 1000 times. The results are shown in Supplementary Figure 7.

Overall, the main conclusions remain unchanged. In particular, TFT (*k92r*) still ranks first, while *k36r* remains the lowest-performing strategy. Although some strategies experience modest rank shifts under probabilistic endings, these changes are relatively small and remain within the range of variation already observed when reproducing the tournament using fixed match lengths.



Supplementary Figure 7: **Probabilistic ending tournament.** Comparison between the original tournament rankings and reproduced rankings under probabilistic and fixed match lengths. Panels a and b show the reproduced ranks against the original ranks for the probabilistic-ending and fixed-length implementations, respectively. Panels c and d show the corresponding rank differences relative to the original tournament. Despite small variations for some strategies, the overall tournament structure and the identities of the highest- and lowest-ranked strategies remain unchanged. Panels b and d are also shown in the main text (Figure 2); however, we reproduce them here to allow for direct comparison with the probabilistic-ending tournament results.

References

- [1] Axelrod, R. More Effective Choice in the Prisoner's Dilemma. *Journal of Conflict Resolution* **24**, 379–403 (1980).
- [2] Stewart, a. J. & Plotkin, J. B. Extortion and cooperation in the Prisoner's Dilemma. *Proceedings of the National Academy of Sciences* **109**, 10134–10135 (2012).
- [3] Guyon, I., Weston, J., Barnhill, S. & Vapnik, V. Gene selection for cancer classification using support vector machines. *Machine learning* **46**, 389–422 (2002).
- [4] Harper, M. *et al.* Reinforcement learning produces dominant strategies for the iterated prisoner's dilemma. *CoRR abs/1707.06307* (2017). URL <http://arxiv.org/abs/1707.06307>.
- [5] The Axelrod project developers. Axelrod-python/axelrod: v4.10.0. <https://doi.org/10.5281/zenodo.3980654>.